

# Consulting Project



The aim of the project is to first identify whether there is a significant difference in features for different species and to come up with a model that can predict the species based on their features.

## Testing difference in features for different species

### Sepal Length

```
data(iris)
model1 = lm(Sepal.Length ~ Species, data = iris)
summary(model1)

##
## Call:
## lm(formula = Sepal.Length ~ Species, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6880 -0.3285 -0.0060  0.3120  1.3120
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.0060     0.0728  68.762 < 2e-16 ***
## Speciesversicolor  0.9300     0.1030   9.033 8.77e-16 ***
## Speciesvirginica   1.5820     0.1030  15.366 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5148 on 147 degrees of freedom
## Multiple R-squared:  0.6187, Adjusted R-squared:  0.6135
## F-statistic: 119.3 on 2 and 147 DF, p-value: < 2.2e-16
```

### Petal Length

```
model2 = lm(Petal.Length ~ Species, data = iris)
summary(model2)
```

```
##
## Call:
```

```
## lm(formula = Petal.Length ~ Species, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.260 -0.258  0.038  0.240  1.348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.46200    0.06086   24.02  <2e-16 ***
## Speciesversicolor  2.79800    0.08607   32.51  <2e-16 ***
## Speciesvirginica   4.09000    0.08607   47.52  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4303 on 147 degrees of freedom
## Multiple R-squared:  0.9414, Adjusted R-squared:  0.9406
## F-statistic: 1180 on 2 and 147 DF, p-value: < 2.2e-16
```

## Petal Width

```
model3 = lm(Petal.Width ~ Species, data = iris)
summary(model3)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.626 -0.126 -0.026  0.154  0.474
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.24600    0.02894    8.50 1.96e-14 ***
## Speciesversicolor  1.08000    0.04093   26.39 < 2e-16 ***
## Speciesvirginica   1.78000    0.04093   43.49 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2047 on 147 degrees of freedom
## Multiple R-squared:  0.9289, Adjusted R-squared:  0.9279
## F-statistic: 960 on 2 and 147 DF, p-value: < 2.2e-16
```

## Principal Component Analysis

As noted from the EDA, the first two principal components capture 95% of the total variability but have significant overlap between versicolor and virginica. We test whether we can come up with a summary statistic of the original data that statistically differ between species.

Using principal component analysis (PCA), we end up with a linear combination of the original variables that capture the most variances among them. The first principal component is approximately proportional

to

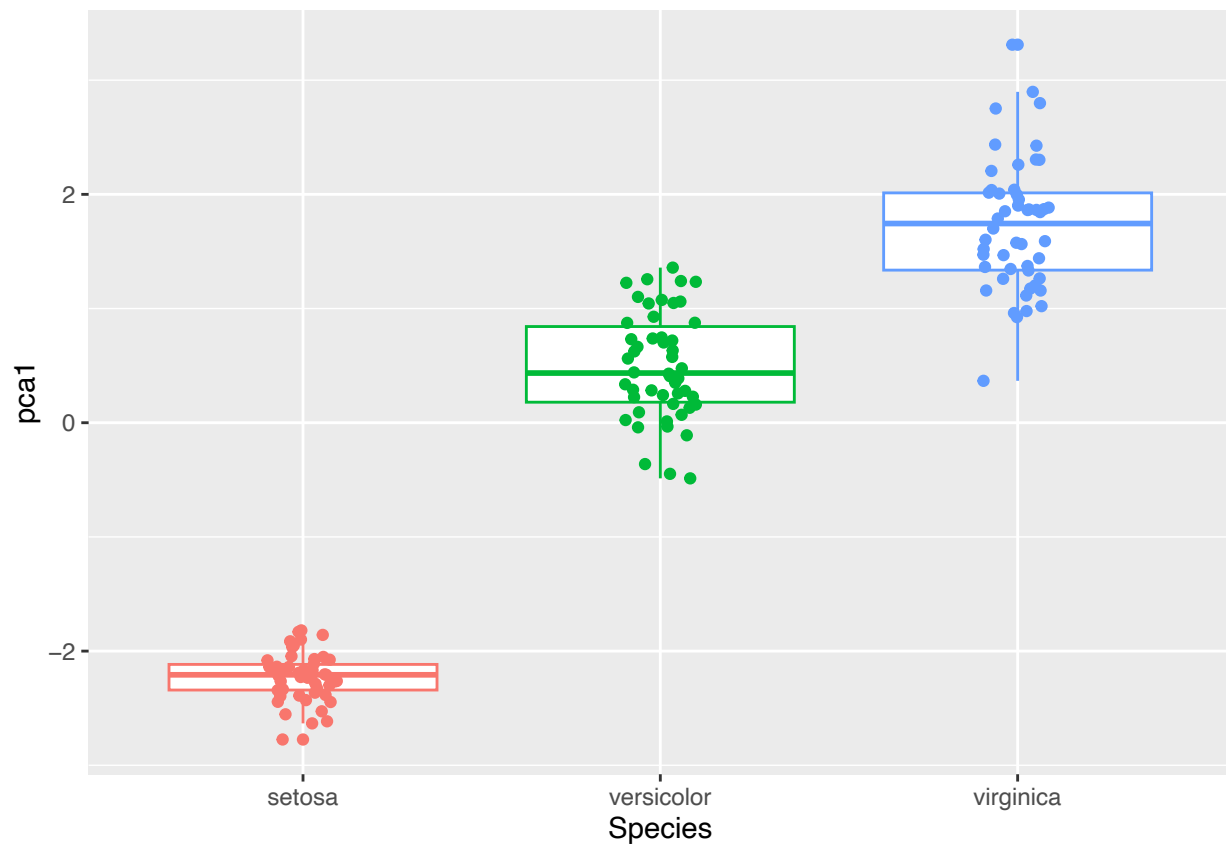
$$SepalLength + PetalLength + PetalWidth - \frac{1}{2}SepalWidth$$

A similar ANOVA test shows that one can distinguish each species based on this first principal component significantly.

```
pca = princomp(iris[,1:4], cor = T)
pca$loadings[,1]
```

```
## Sepal.Length Sepal.Width Petal.Length Petal.Width
## 0.5210659 -0.2693474 0.5804131 0.5648565
```

```
pca1 = pca$scores[,1]
ggplot(iris, aes(x = Species, y= pca1, col = Species)) +
  geom_boxplot() +
  geom_jitter(width = 0.1) +
  theme(legend.position = "none")
```



```
model_pca = lm(pca1 ~ Species, data = iris)
summary(model_pca)
```

```
##
## Call:
## lm(formula = pca1 ~ Species, data = iris)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36129 -0.25971  0.00195  0.23267  1.58239
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.22475    0.06240  -35.65  <2e-16 ***
## Speciesversicolor  2.72120    0.08825   30.83  <2e-16 ***
## Speciesvirginica   3.95306    0.08825   44.79  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4413 on 147 degrees of freedom
## Multiple R-squared:  0.9346, Adjusted R-squared:  0.9337
## F-statistic: 1051 on 2 and 147 DF,  p-value: < 2.2e-16
```

## Classification

### Logistic Regression

```
data(iris)
logistic_model = vglm(Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
                      family = multinomial, data = iris)
predict_probs = predict(logistic_model, type = 'response')
predict_class = colnames(predict_probs)[apply(predict_probs, 1, which.max)]
table(iris$Species, predict_class)
```

```
##              predict_class
##              setosa versicolor virginica
## setosa             50             0             0
## versicolor          0             49             1
## virginica           0             1             49
```

### Training and Test set

```
test_index = seq(1,150, by = 3)
train = iris[-test_index,]
test = iris[test_index,]
logistic_train = vglm(Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
                      family = multinomial, data = train)
predict_probs = predict(logistic_train, newdata = test, type = 'response')
predict_class = colnames(predict_probs)[apply(predict_probs, 1, which.max)]
table(test$Species, predict_class)
```

```
##              predict_class
##              setosa versicolor virginica
## setosa             17             0             0
```

|    |            |   |    |    |
|----|------------|---|----|----|
| ## | versicolor | 0 | 17 | 0  |
| ## | virginica  | 0 | 1  | 15 |